

# Comprehensive Industrial Safety Knowledge Base

## Module I: The Factories Act, 1948 (Safety & Health)

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### Section 21: Fencing of Machinery

Every dangerous part of machinery must be securely fenced [cite: 1]. The Forge-Sense platform must detect moving parts like gears, belts, and shafts in zones tagged 'hazard:moving\_parts' [cite: 1].

### Section 22: Work on Machinery in Motion

Only trained adult workers in tight-fitting clothing may examine machinery in motion [cite: 1]. Forge-Sense must alert on loose clothing or unauthorized workers in these zones [cite: 1].

### Section 36: Dangerous Fumes

Entry into confined spaces with potential fumes requires ventilation and proper breathing apparatus [cite: 1]. Alerts trigger if workers enter without confirmed sensor safety [cite: 1].

## Module II: OISD Guidelines (Oil & Gas)

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### OISD-192: Safety in Construction

Rigid safety protocols for construction sites within hydrocarbon facilities [cite: 2].  
Focuses on permit-to-work systems for high-risk activities [cite: 2].

### OISD-166: Occupational Health

Requires continuous health monitoring and mitigation of exposure to toxic hydrocarbons [cite: 2].

## Module III: DGMS Circulars (Mining)

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### Ventilation Requirements

Underground mines must maintain strict airflow to keep methane and other gases below critical thresholds [cite: 3].

### Ground Control

Mandatory inspection of roof supports and slope stability to prevent structural collapse [cite: 3].

### Machinery Integrity

Periodic testing of winding engines and electrical systems is required for safe mining operations [cite: 3].